Digital ASIC Design Laboratory

Standard Cell Logic Synthesis Specification

Target Design: **i2c_master** | EDA Tool: **Synopsys Design Compiler**

1. Project Scope & Design Objectives

The primary objective of this laboratory assignment is to execute automated logic synthesis on the top-level `i2c_master` Verilog design using Synopsys `dc_shell`. Map the unmapped RTL to the designated 90nm standard cell technology library under worst-case operating conditions.

- **Target Library File:** `std_cells/saed90nm_max_hvt.db`
- **Process Corner:** Slow / Worst-Case Corner Analysis

2. Environmental & Timing Constraints Script (`cons.tcl`)

Author a comprehensive Tcl constraint file named `cons.tcl` incorporating the following specifications:

2.1 Clock & Uncertainty Specifications

- **Primary Clock Period:** Define system clock target to **8.0 ns**.
- **Setup Timing Margin:** Apply a setup clock uncertainty of **0.24 ns** (3% of T_{clk}).
- **Hold Timing Margin:** Apply a hold clock uncertainty of **0.16 ns** (2% of T_{clk}).

2.2 Boundary Port Constraints

- **Input Delay:** Constrain all input signals (excluding `CLK` and `RST`) to **1.6 ns** (20% of T_{clk}).
- **Output Delay:** Constrain all output ports to **1.6 ns** (20% of T_{clk}).
- **Port Driving Cell:** Set input driver to cell `NBUFFX2_HVT` across all input ports.
- **Output Loading:** Apply a capacitive load of **100 fF** on all output signal lines.

2.3 Network Attributes & Wire Load Models

- **Ideal Networks:** Attach the `ideal_network` attribute to `CLK` and `RST` ports to prevent clock tree buffer insertion during early synthesis logic optimization.
- **Operating Environment:** Explicitly configure the slow corner operating condition and its associated wire load model from `saed90nm_max_hvt`.

3. Diagnostic Reporting Requirements

Your synthesis execution script must generate and export the following diagnostic evaluation reports:

Report Category	Design Compiler Command	Target Output File
Power Analysis	`report_power -hierarchy`	`power.rpt`
Area Utilization	`report_area -hierarchy`	`area.rpt`
Setup Timing Analysis	`report_timing -max_paths 10 -delay_type max`	`setup.rpt`
Hold Timing Analysis	`report_timing -max_paths 10 -delay_type min`	`hold.rpt`
Clock Attributes	`report_clock -attributes`	`clocks.rpt`
Constraint Violations	`report_constraint -all_violators`	`constraints.rpt`

4. Required Submission Deliverables

Submit the complete synthesis package containing all generated files listed below:

Category	Required Artifact Filename
Synthesis Tcl Scripts	`syn_script.tcl`, `cons.tcl`
Mapped Gate-Level Netlist	`i2c_master.v`
Design Database & Constraints	`i2c_master.ddc`, `i2c_master.sdc`
Back-Annotation Timing File	`i2c_master.sdf`
Synthesis Diagnostic Reports	`setup.rpt`, `hold.rpt`, `clocks.rpt`, `constraints.rpt`, `area.rpt`, `power.rpt`